Flutter Local DB and Persistence

1. Explain the difference between local storage options (shared_preferences, SQLite, Hive).

Ans : Flutter provides different local storage options to store data on the device. The most commonly used options are **Shared Preferences**, **SQLite**, and **Hive**. Each one is used for different purposes depending on the type and size of data.

1. Shared Preferences

Shared Preferences is a simple key-value storage system. It is mainly used to store small amounts of data such as user settings, login status, theme mode, language preferences, etc.

Advantages:

- Easy to use
- Fast for small data
- Good for storing simple values like String, int, bool

Disadvantages:

- Not suitable for large data
- Cannot store complex structured data easily

2. SQLite

SQLite is a relational database used to store structured data in tables with rows and columns. It is useful when the app needs CRUD operations (Create, Read, Update, Delete).

Advantages:

- Best for large and structured data
- Supports SQL queries
- Good for search, filter, sort, and relationships between data

Disadvantages:

- More complex than Shared Preferences
- Requires database setup and SQL knowledge

3. Hive

Hive is a lightweight and fast NoSQL database for Flutter. It stores data in boxes and is commonly used for offline apps and storing objects.

Advantages:

- Very fast performance
- Easy to use
- Can store custom objects
- No SQL required

Disadvantages:

- Limited complex query support
- Not ideal for relational data

2. Describe CRUD operations and how they are implemented in SQLite or Hive.

Ans: CRUD stands for **Create, Read, Update, and Delete**. These are the four basic operations used to manage data in local databases such as SQLite or Hive.

1. Create

Create means adding new data into the database.

Example: Adding a new note, task, or user record.

- In **SQLite**, data is inserted using the INSERT query.
- In **Hive**, data is added using box.add() or box.put().

2. Read

Read means retrieving stored data from the database.

Example: Displaying all saved notes or tasks.

- In **SQLite**, data is fetched using the SELECT query.
- In **Hive**, data is accessed using box.values or box.get().

3. Update

Update means modifying existing data.

Example: Editing a note or changing user information.

- In **SQLite**, data is updated using the UPDATE query.
- In **Hive**, data is updated using box.put() with the same key.

4. Delete

Delete means removing data from the database.

Example: Deleting a completed task or note.

- In **SQLite**, data is deleted using the DELETE query.
- In **Hive**, data is removed using box.delete().

CRUD in SQLite

SQLite uses tables with rows and columns. It is useful for structured data and supports SQL commands like INSERT, SELECT, UPDATE, and DELETE.

CRUD in Hive

Hive stores data in boxes. It is simple, fast, and does not require SQL queries. It is commonly used in Flutter for local storage.

3. Explain the advantages and use cases for shared_preferences.

Ans: Shared Preferences is a local storage option in Flutter used to store small amounts of data in the form of key-value pairs. It is mainly used for saving simple app data on the device.

Advantages of Shared Preferences

1. **Easy to Use**
   Shared Preferences has a simple API, making it easy for beginners to implement.
2. **Fast Performance**
   It is quick for storing and retrieving small data such as strings, integers, booleans, and doubles.
3. **Persistent Storage**
   Data remains saved even after the app is closed or restarted.
4. **Lightweight Storage**
   It does not require database setup or table creation.
5. **Useful for Simple Data**
   Best for storing settings and preferences.

Use Cases of Shared Preferences

1. **Login Status**
   Save whether the user is logged in or not.
2. **Theme Settings**
   Store dark mode or light mode preference.

3. **Language Preference**
   Save selected language of the app.
4. **User Name or Email**
   Store basic user information.
5. **Onboarding Screen Status**
   Save whether the user has already seen the intro screens.
6. **Small App Settings**
   Save notification settings, sound settings, etc.